

Commonsense-Driven Symbolic ReAct-NLI Prompting (CSR-NLI)

System Prompt

You are an advanced AI assistant that uses **Neuro Symbolic AI (NSAI)** principles to classify employee conversational messages. Your task involves:

Framework:

- **Natural Language Inference (NLI):**
 - **Premise:** The employee's conversational message.
 - **Hypothesis:** A single commonsense explanation linking the message to one predefined category.
 - **Label:**
 - **Entailment:** The reasoning aligns with the hypothesis.
 - **Contradiction:** The reasoning conflicts with the hypothesis.
 - **Neutral:** The reasoning provides insufficient evidence.

Categories:

Classify messages into one of the following categories:

1. **C1:** No reason
2. **C2:** Jobs and careers
3. **C3:** Medication
4. **C4:** Relationship
5. **C5:** Alienation
6. **C6:** Bias or abuse

Process:

1. **Generate a Commonsense Hypothesis:**
 - Create a single commonsense-based hypothesis explaining the premise and linking it to one category.
 - Example: For "I am anxious because of a deadline," generate the hypothesis: "Deadlines often cause stress in jobs and careers."
2. **Iterative Reasoning Using ReAct:**
 - Perform iterative reasoning using the following structure:
 - **Thought:** Analyze the premise and propose reasoning.
 - **Action:** Compare the reasoning to the hypothesis.
 - **Observation:** Determine if the reasoning aligns with the hypothesis.

- Continue refining until **Entailment** is achieved.
- 3. **Label and Finalize:**
 - Assign a label (**Entailment**, **Contradiction**, or **Neutral**) to each iteration of reasoning.
 - Finalize the reasoning and category once **Entailment** is achieved.
- 4. **Provide Explanation:**
 - Output the reasoning for the selected category, ensuring alignment with the commonsense hypothesis.

Output Format:

- **Reasoning:** Explain why the category was chosen, ensuring alignment with the hypothesis.
 - **Category:** The selected category.
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User Prompt

Premise: "I am feeling anxious because I am not sure if I'll meet the deadline at work."

Example Workflow Demonstrating ReAct Iterations

Step 1: Generate Commonsense Hypothesis

- **Hypothesis:** "Deadlines often cause stress in jobs and careers."
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Step 2: Iterative Reasoning Using ReAct

1. Iteration 1:

Thought 1: "The message mentions anxiety, but it doesn't explicitly link it to work. Perhaps the anxiety stems from another source."

- **Action 1:** "Validate reasoning against the hypothesis: 'Deadlines often cause stress in jobs and careers.'"
- **Observation 1:** "The hypothesis suggests a job-related cause, but the reasoning doesn't align fully."
- **Label:** **Neutral**.

2. Iteration 2:

- **Thought 2:** "The message mentions a 'deadline,' which might imply work-related stress. However, it could also be personal."
- **Action 2:** "Reassess reasoning against the hypothesis: 'Deadlines often cause stress in jobs and careers.'"
- **Observation 2:** "The hypothesis aligns partially, but more evidence is needed to confirm a job connection."

- **Label: Neutral.**
 - 3. **Iteration 3:**
 - **Thought 3:** "The message explicitly says 'deadline at work,' linking the anxiety directly to a job-related cause."
 - **Action 3:** "Validate reasoning against the hypothesis: 'Deadlines often cause stress in jobs and careers.'"
 - **Observation 3:** "The hypothesis aligns perfectly with the message context."
 - **Label:** Entailment.
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Step 3: Final Reasoning and Output

- **Reasoning:** "The message explicitly mentions 'deadline at work,' linking anxiety to job-related stress. This aligns with the hypothesis that deadlines often cause stress in jobs."
 - **Category:** C2 (Jobs and careers)
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Generalized Prompt

Premise: "I am feeling anxious because I am not sure if I'll meet the deadline at work."

Thought 1: "The message mentions anxiety but does not specify if it is work-related."

Action 1: Validate this reasoning against the commonsense hypothesis: "Deadlines often cause stress in jobs and careers."

Observation 1: The reasoning partially aligns but lacks clarity.

Label 1: Neutral.

Thought 2: "The message mentions a 'deadline,' which might imply job-related stress."

Action 2: Validate reasoning against the hypothesis.

Observation 2: The hypothesis aligns partially.

Label 2: Neutral.

Thought 3: "The message explicitly says 'deadline at work,' which directly links anxiety to a job-related cause."

Action 3: Validate reasoning against the hypothesis.

Observation 3: The hypothesis fully aligns.

Label 3: Entailment.

Final Output:

Reasoning: "The message explicitly says 'deadline at work,' which directly links anxiety to a job-related cause."

Category: C2 (Jobs and careers)

Background of Commonsense-Driven Symbolic ReAct-NLI (CSR-NLI)

The **Commonsense-Driven Symbolic ReAct-NLI (CSR-NLI)** method is an advanced framework for classifying conversational messages by integrating **Natural Language Inference (NLI)**, **commonsense reasoning**, **symbolic logic**, and the **ReAct framework**. This method builds upon the principles of **Neuro Symbolic AI (NSAI)** to achieve a dynamic, iterative, and interpretable approach to reasoning and classification.

Key Features of CSR-NLI

Clarity and Usability:

- Consolidates category-specific instructions into the system prompt, making the user prompt concise and task-specific.
- Dynamically generates a **single commonsense hypothesis** using the OpenAI LLM, avoiding the complexity of predefined hypotheses.
- The user only provides the input message (Premise), ensuring simplicity and adaptability across tasks.

Alignment with NSAI and NLI:

- Adheres to NLI principles by:
 - Treating the input message as the **Premise**.
 - Dynamically generating a **Commonsense Hypothesis** that serves as a benchmark for reasoning.
 - Iteratively assigning labels (**Entailment**, **Contradiction**, **Neutral**) to generated reasoning based on alignment with the commonsense hypothesis.
- Embeds symbolic reasoning by validating neural reasoning outputs against commonsense benchmarks.

Integration of Advanced Reasoning Frameworks:

1. **Commonsense Reasoning:**
 - Dynamically generates commonsense hypotheses using OpenAI LLM.
 - Validates generated reasoning against the hypothesis to ensure alignment with real-world knowledge, enhancing robustness and accuracy.
2. **ReAct Framework:**
 - Uses an iterative cycle of **Thought**, **Action**, and **Observation** to refine reasoning until alignment with the hypothesis is achieved.
 - Example:
 - **Thought:** Analyze the premise and propose reasoning.
 - **Action:** Compare reasoning to the commonsense hypothesis.
 - **Observation:** Evaluate alignment and iterate if necessary.
3. **Dynamic NLI:**

- Generates and refines reasoning dynamically for each input, ensuring adaptability to diverse conversational contexts.
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Iterative Refinement for Robustness

- Employs a **feedback loop** that refines reasoning until the generated reasoning aligns with the commonsense hypothesis.
 - Reduces reliance on static rules or assumptions, making the framework resilient to edge cases or ambiguous inputs.
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Transparency and Explainability

- Explicitly outlines each reasoning step, ensuring interpretability and traceability of decisions.
 - Final classifications are justified with clear, symbolic reasoning, providing trust and transparency in the system.
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Integration of Advanced Prompt Techniques

ReAct-Inspired Iterative Framework:

- Dynamically interleaves reasoning and actions:
 - **Thought:** Analyze and generate intermediate reasoning.
 - **Action:** Validate reasoning against the commonsense hypothesis.
 - **Observation:** Refine reasoning and iterate as needed.
- Balances flexibility and interpretability for complex reasoning tasks.

Commonsense Validation:

- Treats the commonsense hypothesis as a benchmark to validate generated reasoning.
- Ensures that predictions align with real-world knowledge, reducing inconsistencies.

Chain-of-Thought (CoT):

- Breaks down complex reasoning tasks into intermediate steps, improving interpretability and ensuring logical consistency.

Few-Shot and Meta Prompting:

- Demonstrates expected behavior through examples embedded in the system prompt.
 - Structures outputs for ease of interpretation and reuse across tasks.
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Flexibility and Extensibility

Dynamic Hypothesis Generation:

- Adapts to varied conversational inputs without requiring pre-defined hypotheses.
- Handles nuanced and complex reasoning tasks efficiently.

Generalizable Framework:

- Suitable for diverse applications requiring reasoning and classification, including:
 - Emotion analysis.
 - Contextual interpretation.
 - Decision-making processes.

Scalability:

- The lightweight architecture supports seamless scaling across tasks and categories without significant reconfiguration.
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Key Advantages

1. **Accuracy:**
 - Iterative refinement ensures that the final reasoning and classification align precisely with the input message and commonsense hypothesis.
 2. **Interpretability:**
 - Clear reasoning steps, labels, and intermediate observations enhance transparency.
 3. **Robustness:**
 - Handles ambiguity and diverse conversational contexts effectively by leveraging dynamic reasoning and commonsense validation.
 4. **Efficiency:**
 - Combines the strengths of neural reasoning and symbolic validation in a lightweight, scalable system.
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Conclusion

The **CSR-NLI** framework achieves a seamless integration of commonsense reasoning, symbolic logic, and iterative refinement through the ReAct framework. It strikes an optimal balance between theoretical rigor, practical utility, and simplicity, creating a robust, transparent, and scalable method for classifying conversational messages. This framework is adaptable for diverse tasks, ensuring its relevance and applicability across domains.